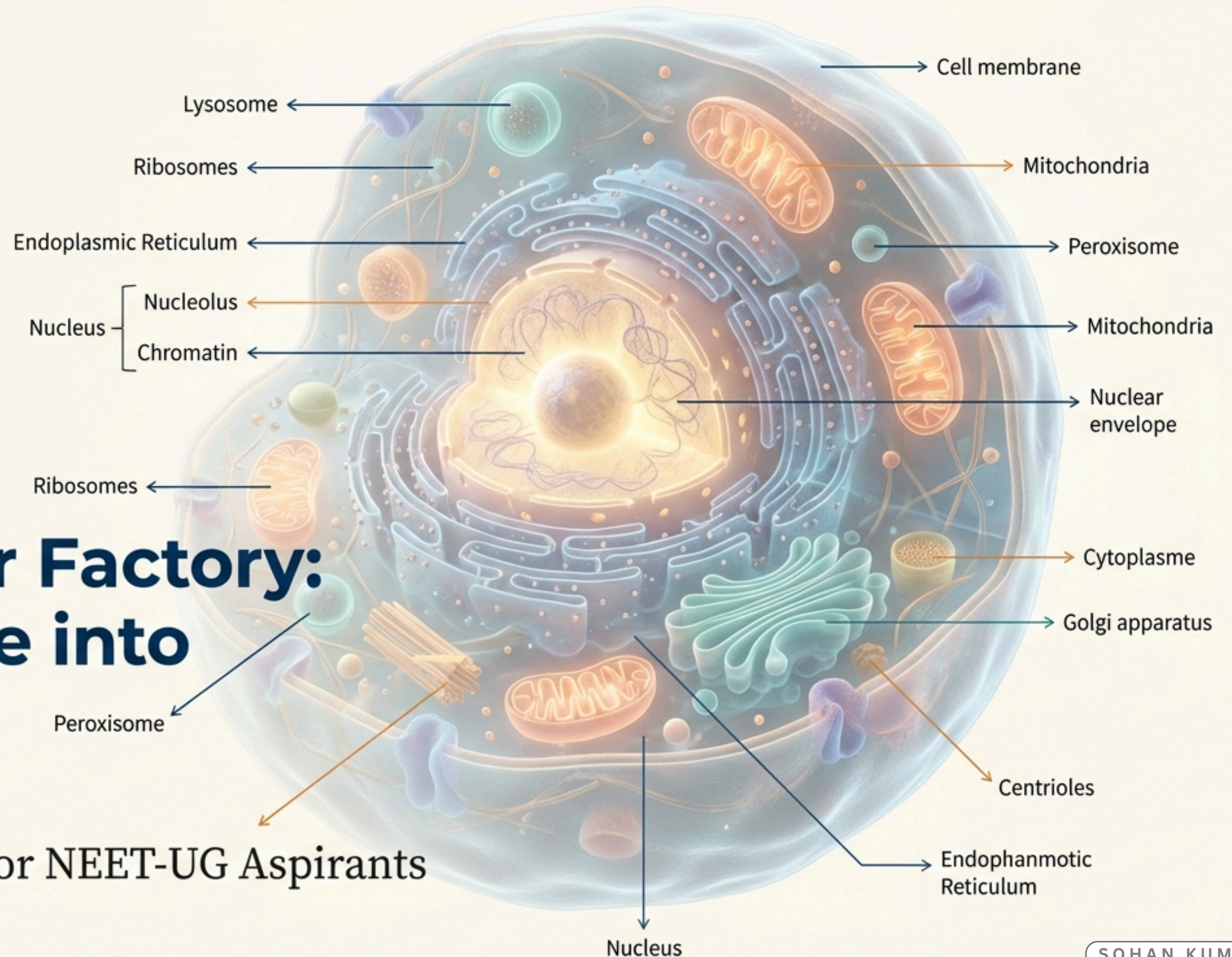


The Cellular Factory: A Deep Dive into Eukaryotic Organelles

An Essential Guide for NEET-UG Aspirants



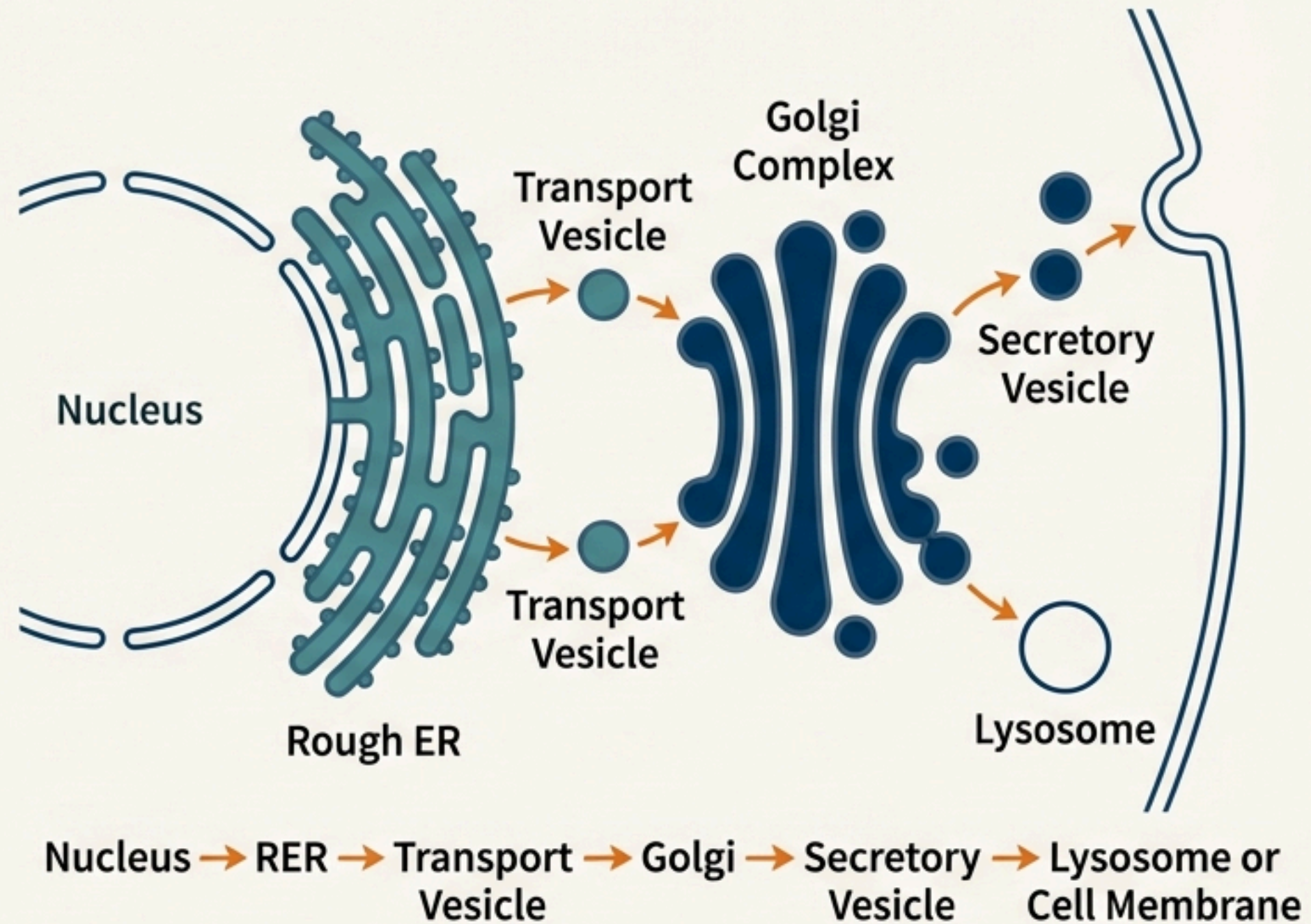
The Endomembrane System: A Coordinated Production Network

Many membranous organelles function together as a single, coordinated unit called the **endomembrane system**.

This system is responsible for protein and lipid synthesis, modification, packaging, and transport.

Members of the System: Endoplasmic Reticulum (ER), Golgi Complex, Lysosomes, Vacuoles.

KEY CONCEPT FOR NEET: Mitochondria, Chloroplasts, and Peroxisomes are **NOT** considered part of the endomembrane system because their functions are not coordinated with it.



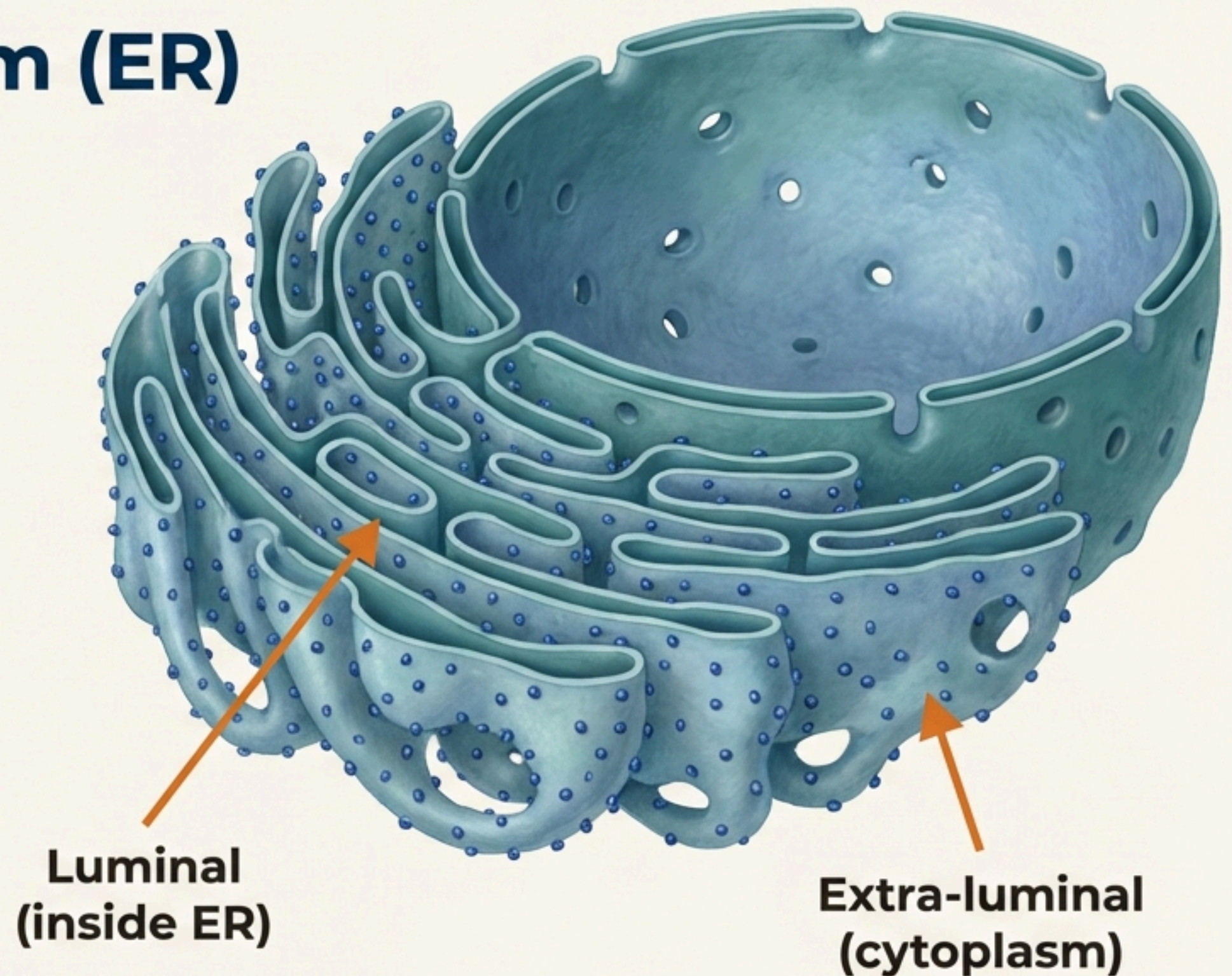
The Assembly Line: Endoplasmic Reticulum (ER)

The ER is a network (reticulum) of tiny tubular structures scattered throughout the cytoplasm. It divides the intracellular space into two distinct compartments:

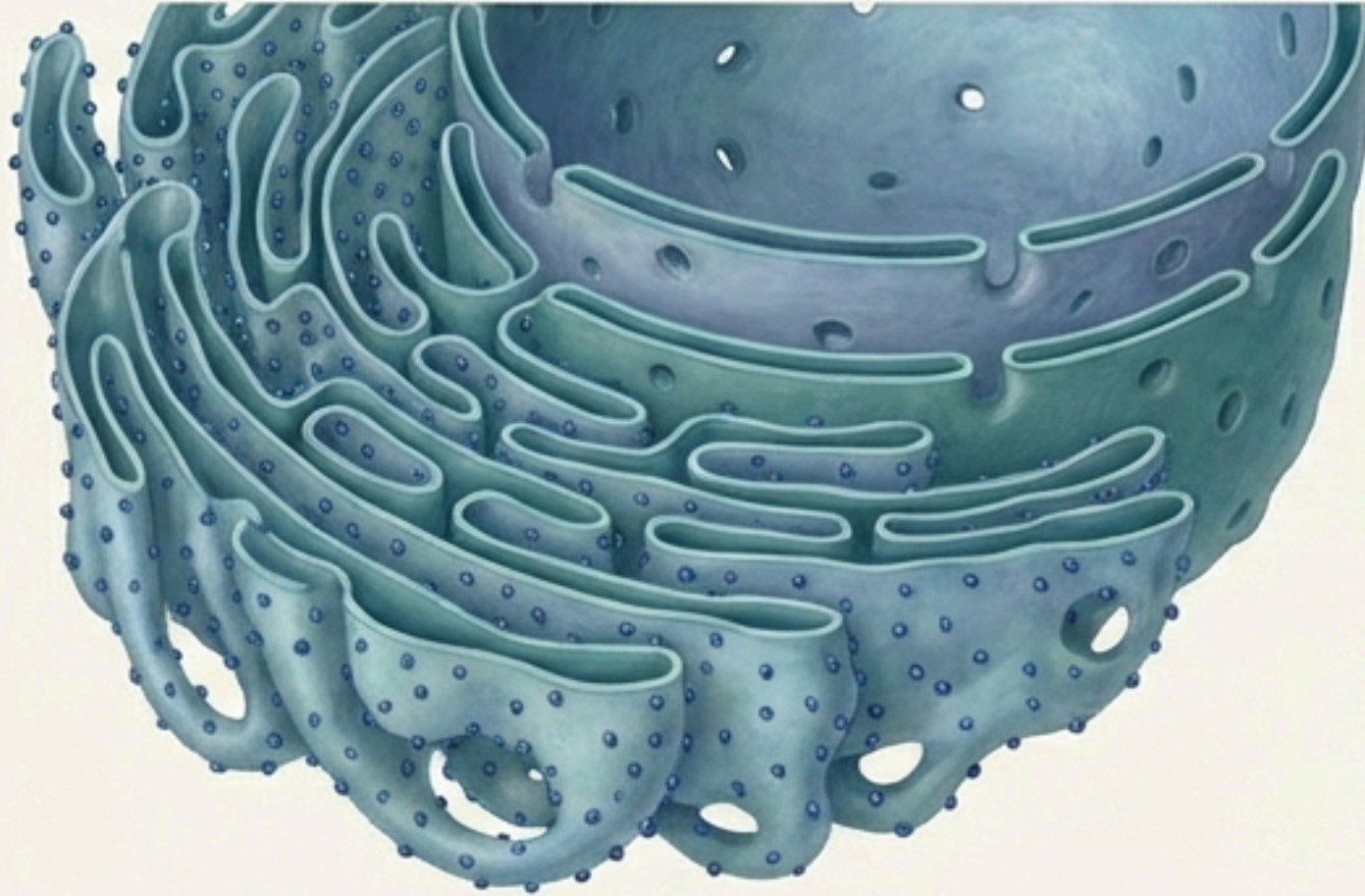
Luminal: The space inside the ER network.

Extra-luminal: The cytoplasm surrounding the ER.

The ER exists in two interconnected forms: Rough ER (RER) and Smooth ER (SER).

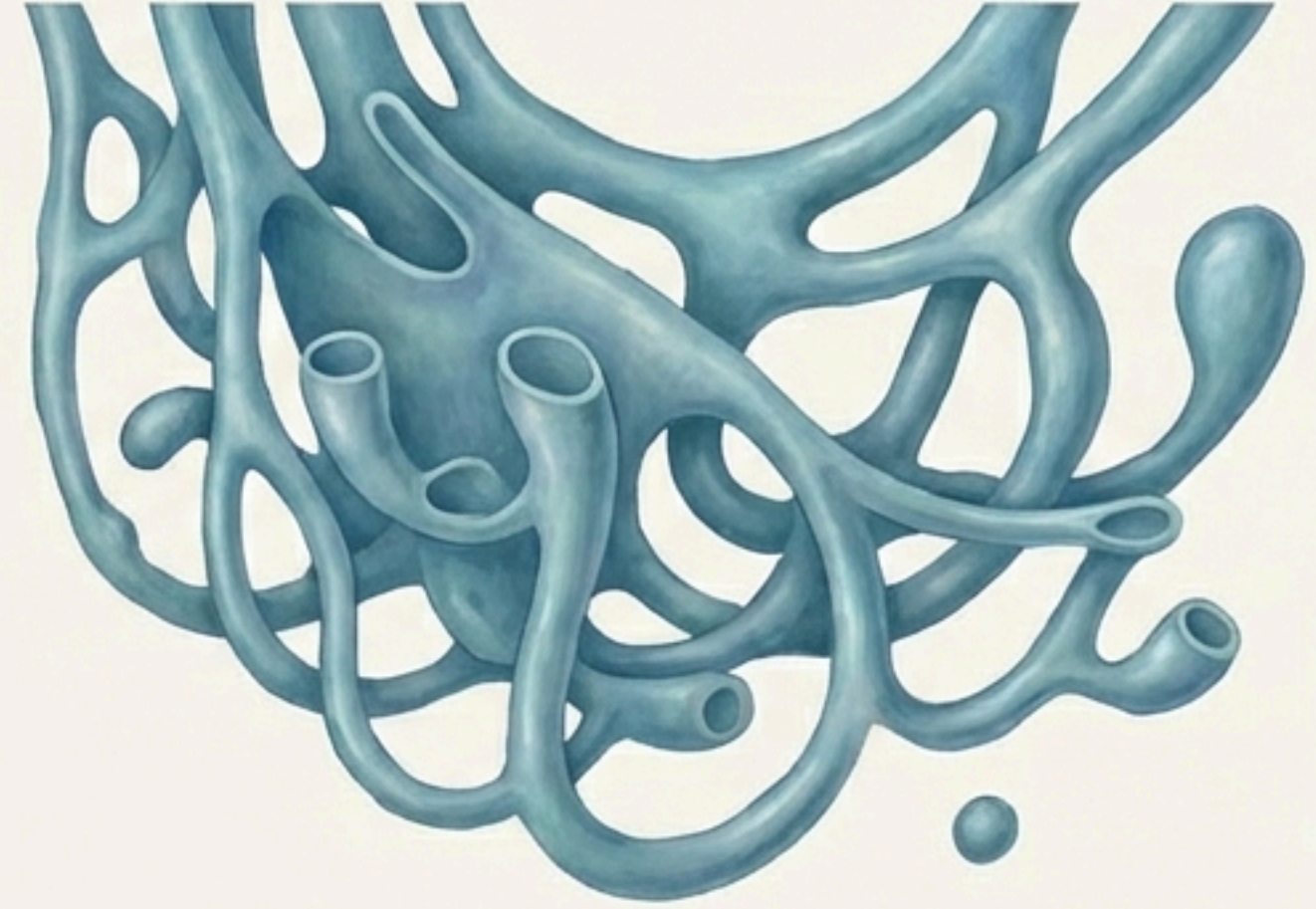


RER vs. SER: Specialized Functions



Rough Endoplasmic Reticulum (RER)

- Bears **ribosomes** on its outer surface.
- Frequently found in cells actively involved in **protein synthesis and secretion**.
- Extensive and continuous with the outer membrane of the nucleus.



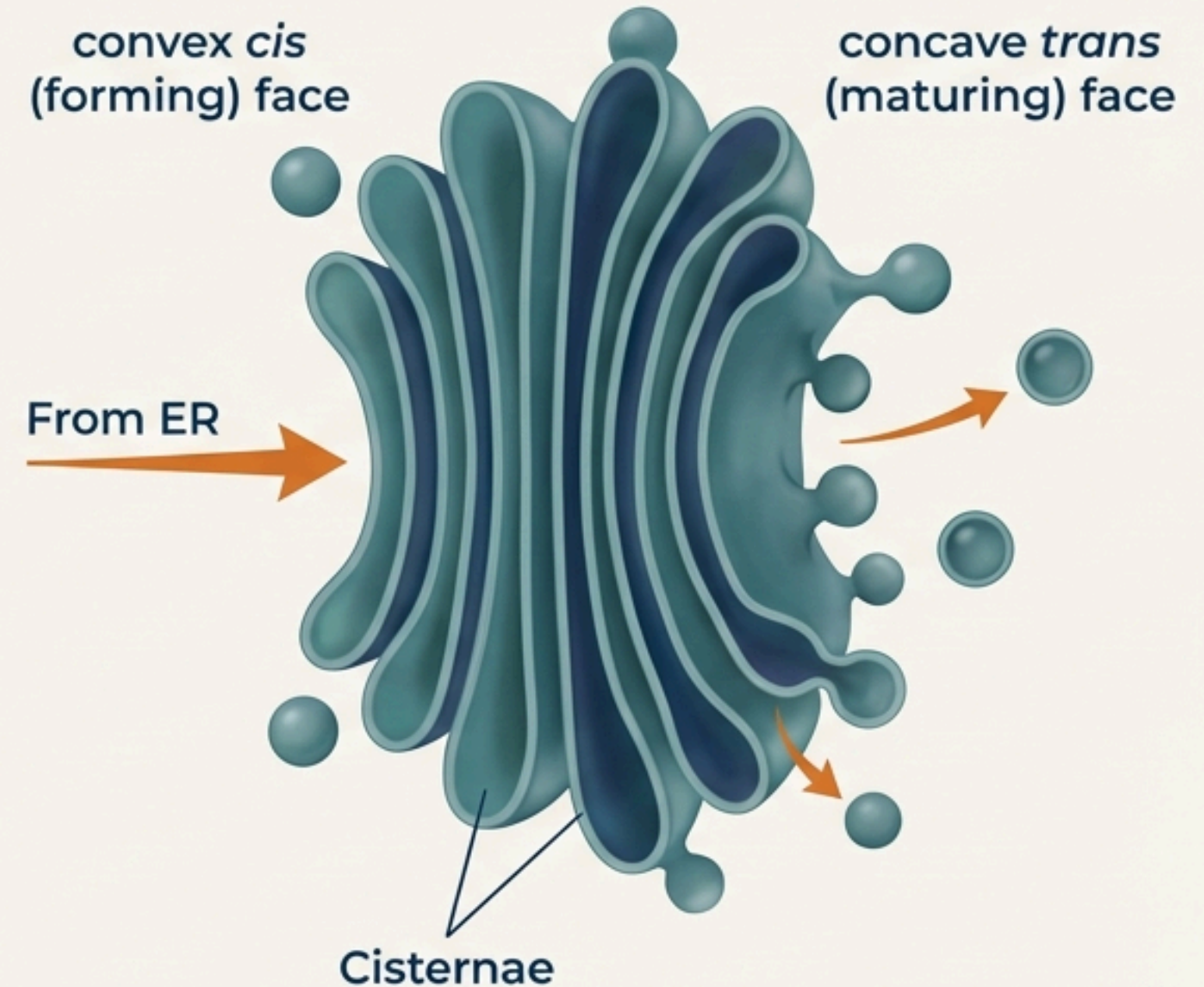
Smooth Endoplasmic Reticulum (SER)

- Appears smooth due to the **absence of ribosomes**.
- The major site for the **synthesis of lipid**.

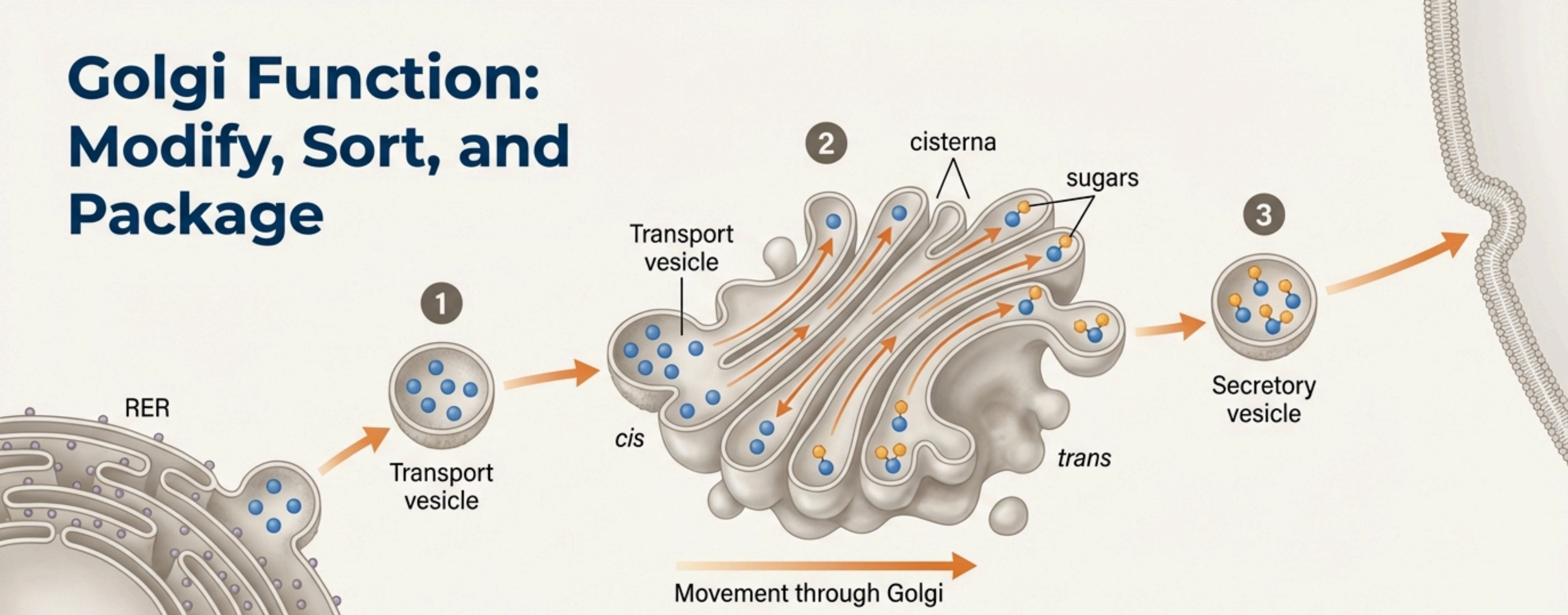
KEY CONCEPT FOR NEET: In animal cells, **lipid-like steroidal hormones** are synthesized in the SER.

The Finishing & Shipping Dept: Golgi Apparatus

- First observed by **Camillo Golgi** in 1898.
- **Structure:** Consists of many flat, disc-shaped sacs called **cisternae** (0.5µm to 1.0µm diameter) stacked parallel to each other.
- **Distinct Polarity:** The Golgi cisternae are concentrically arranged with two distinct faces:
 - **cis face (forming face):** Convex side, oriented towards the nucleus, receives vesicles from the ER.
 - **trans face (maturing face):** Concave side, oriented towards the plasma membrane, releases vesicles.



Golgi Function: Modify, Sort, and Package



- **Principal Function:** Packaging materials for delivery, either to intracellular targets or for secretion outside the cell. This explains its close association with the ER.

- Materials from the ER fuse with the *cis* face and move towards the *trans* face.
- Proteins synthesized on the RER are **modified** in the Golgi cisternae.

KEY CONCEPT FOR NEET: The Golgi apparatus is the important site of formation of **glycoproteins and glycolipids**.

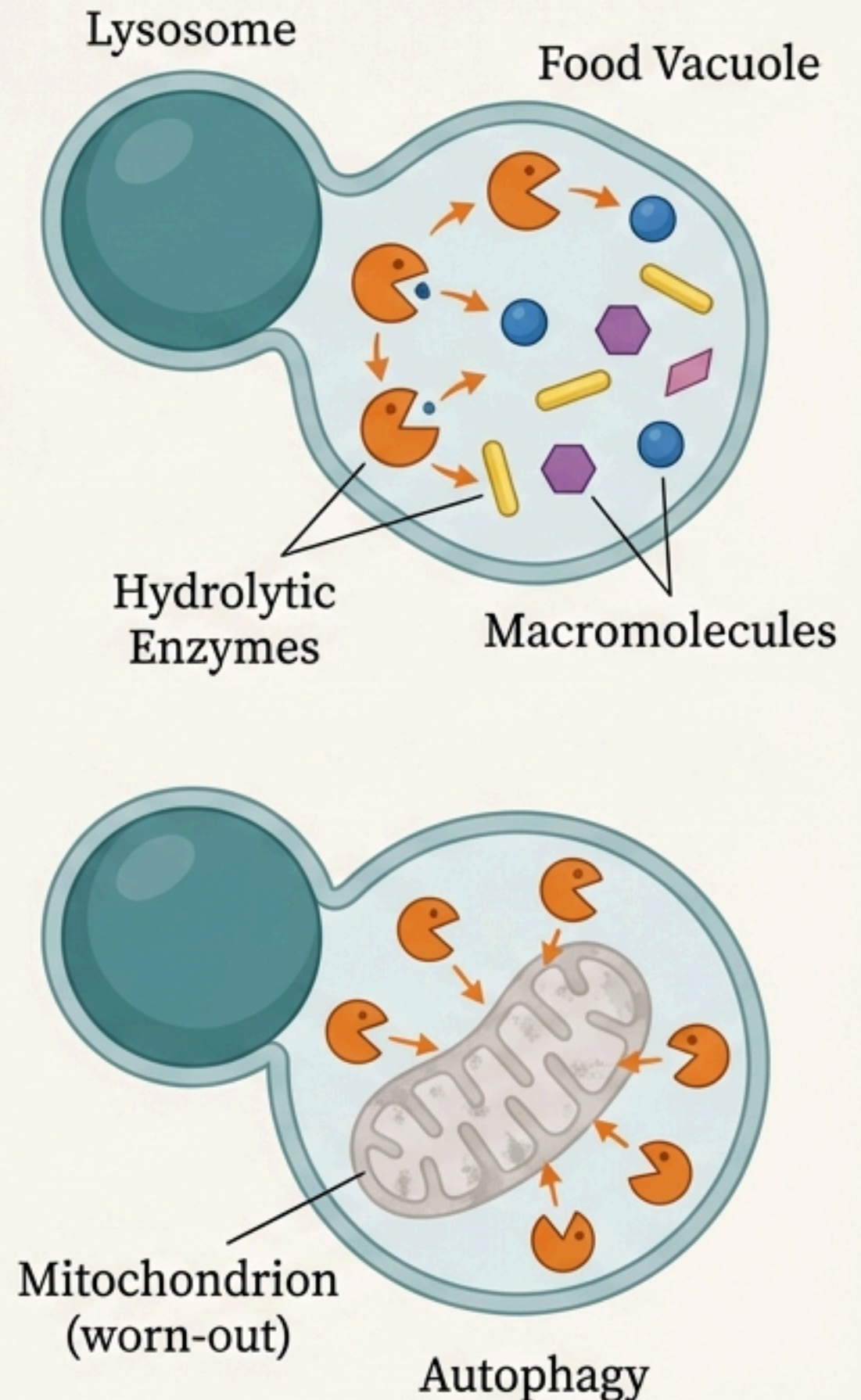
The Recycling Center: Lysosomes

- Membrane-bound vesicular structures formed by the process of packaging in the Golgi apparatus.
- These enzymes are capable of digesting carbohydrates, proteins, lipids, and nucleic acids and are optimally active at an **acidic pH**.

KEY CONCEPT FOR NEET:

Lysosomes are rich in almost all types of **hydrolytic enzymes** (hydrolases), including:

- Lipases (digest lipids)
- Proteases (digest proteins)
- Carbohydrases (digest carbohydrates)

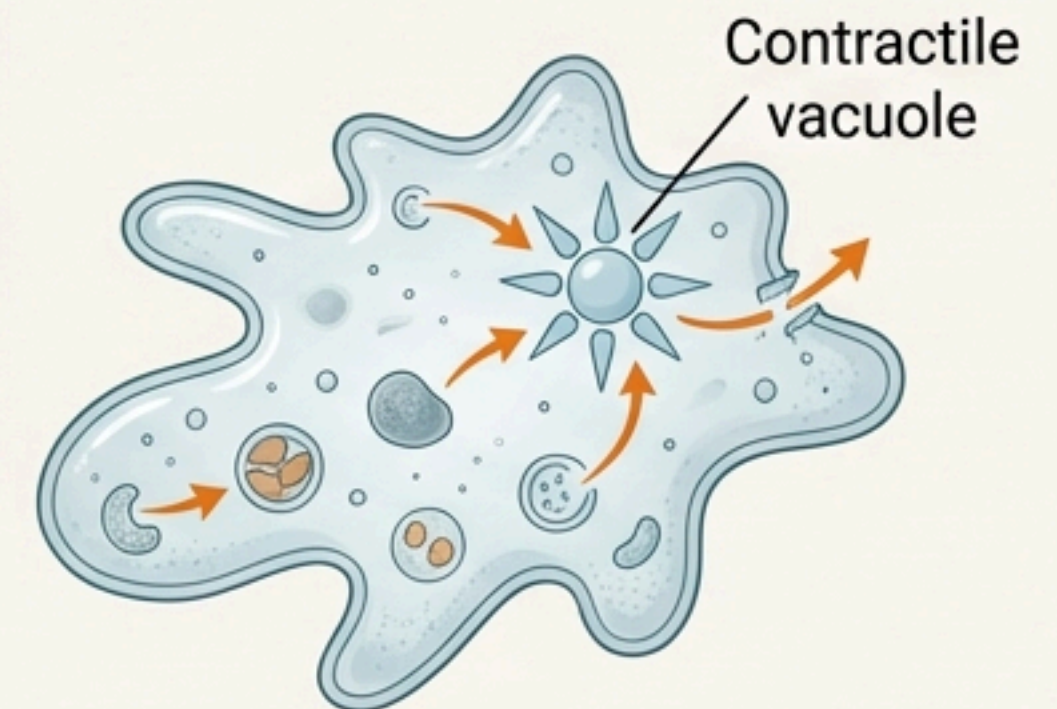
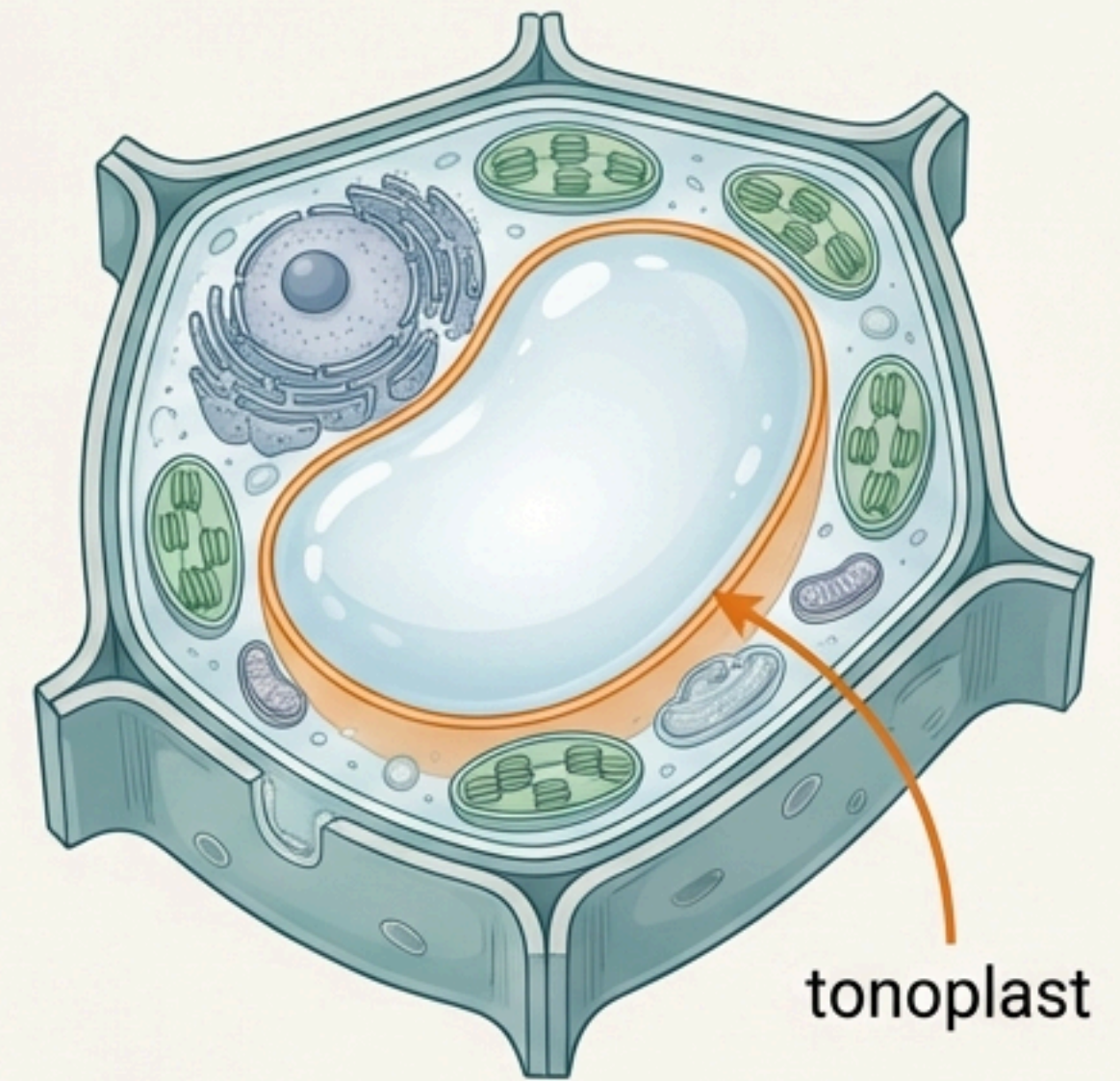


The Storage Warehouse: Vacuoles

- A membrane-bound space in the cytoplasm containing water, sap, excretory products, and other materials.
- The vacuole's single membrane is called the **tonoplast**.
- **In Plant Cells:** Can occupy up to **90% of the cell volume**.

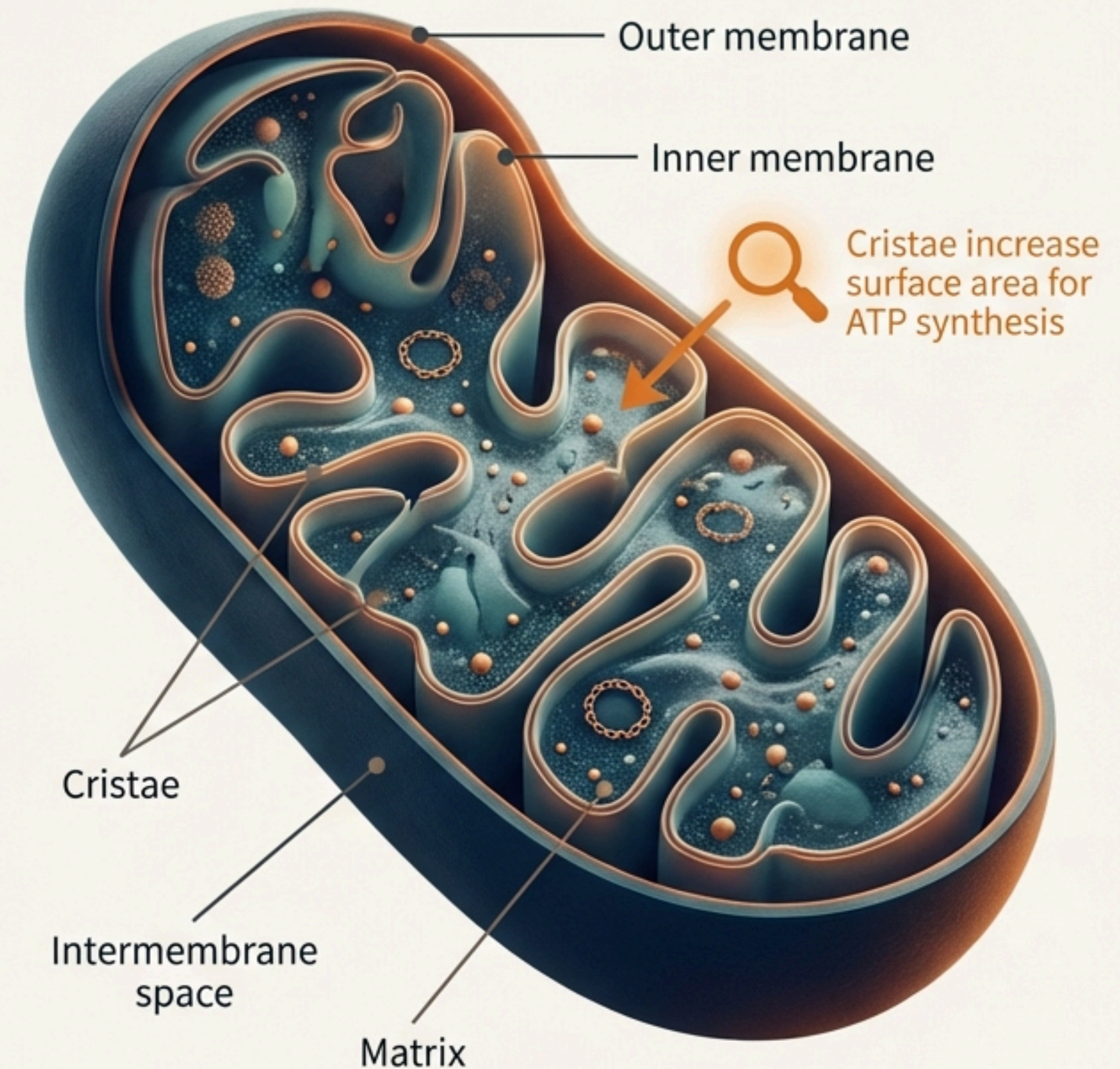
KEY CONCEPT FOR NEET: In plants, the tonoplast facilitates the transport of ions and other materials **against concentration gradients** into the vacuole.

- **Specialized Vacuoles:**
 - **Contractile vacuole** in *Amoeba* is crucial for osmoregulation and excretion.
 - **Food vacuoles** are formed in protists by engulfing food particles.



The Powerhouse: Mitochondria

- Typically sausage-shaped or cylindrical organelles.
- Known as the '**powerhouses**' of the cell because they are the sites of **aerobic respiration** and produce cellular energy in the form of **ATP**.
- **Double Membrane Structure:**
 - A smooth, continuous limiting boundary.
 - Forms numerous infoldings called **cristae**, which significantly **increase the surface area**.
- The inner compartment is filled with a dense fluid called the **matrix**.



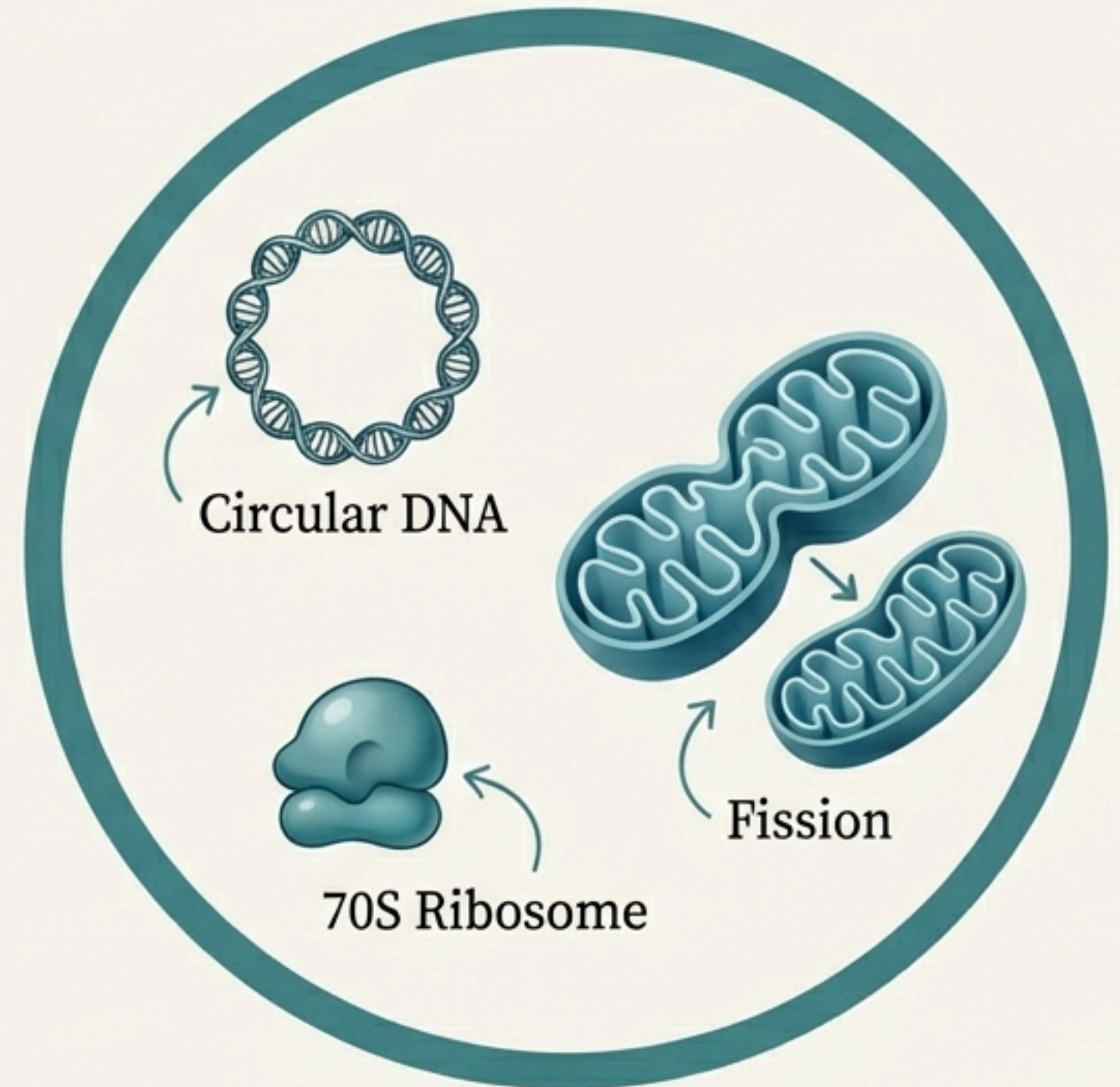
Mitochondria: A Semi-Autonomous Organelle

- Mitochondria are considered **semi-autonomous** organelles.
- Mitochondria divide by **fission**.

KEY CONCEPT FOR NEET:

The matrix possesses its own:

- Single, **circular DNA** molecule
- A few **RNA** molecules
- **Ribosomes** (70S type), which are the same type found in prokaryotes.
- Components required for **protein synthesis**.



The Solar Farm & Storage: Plastids

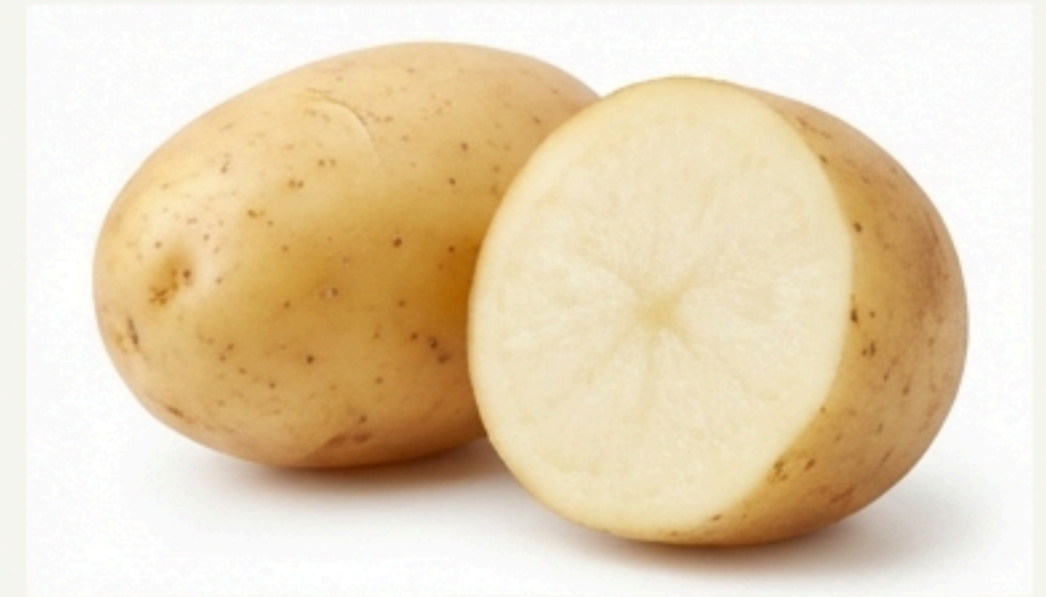
- Found in all plant cells and in euglenoides.
- They bear specific pigments, imparting specific colours to plants.



Chloroplasts



Chromoplasts



Leucoplasts (Amyloplasts)

Classification based on Pigment Type:

Chloroplasts:

Contain chlorophyll and carotenoids for photosynthesis.

Chromoplasts:

Contain fat-soluble carotenoids (carotene, xanthophylls) giving yellow, orange, or red colors.

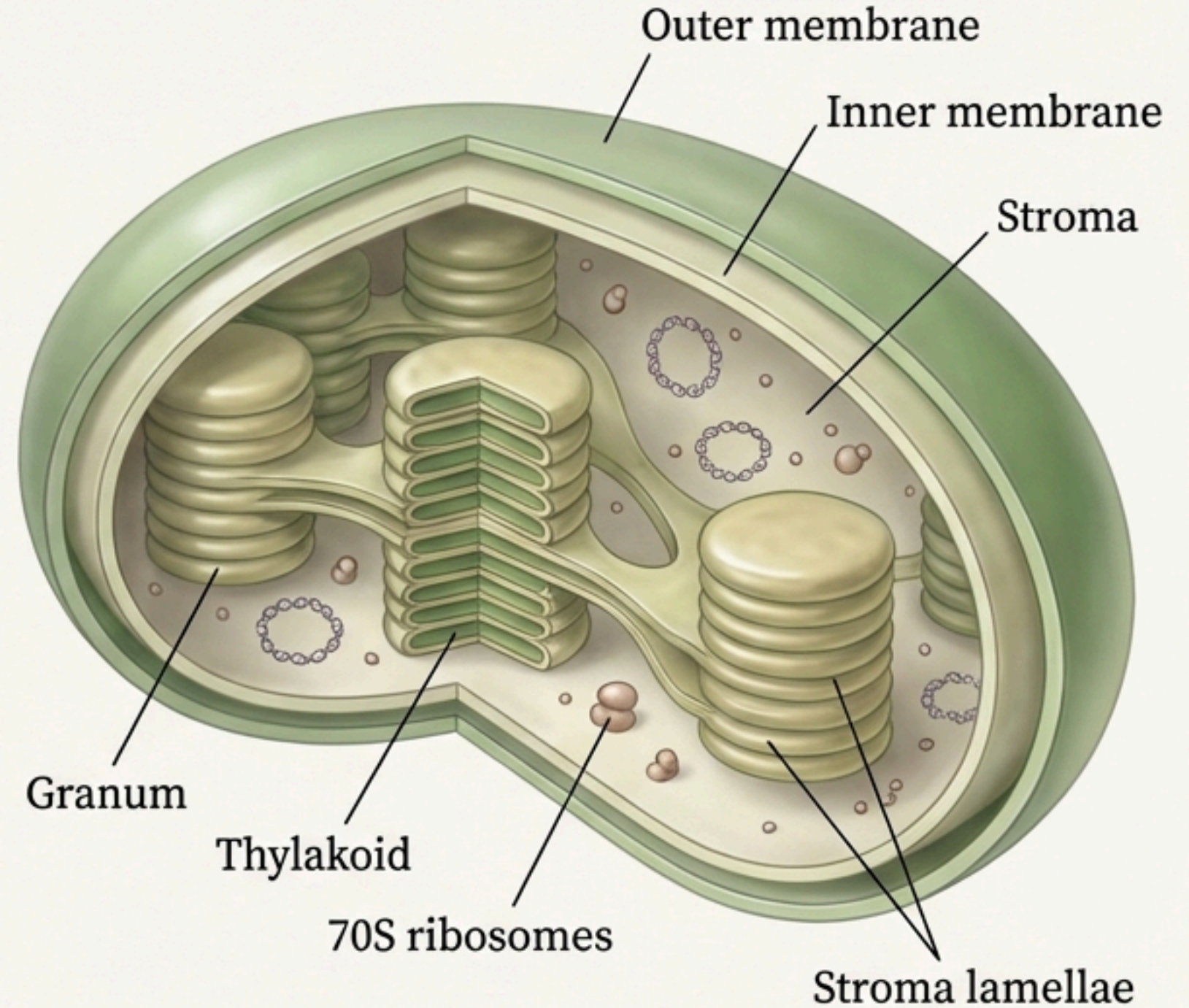
Leucoplasts:

Colorless plastids for nutrient storage (e.g., Amyloplasts store starch, Elaioplasts store oils, Aleuroplasts store proteins).

Anatomy of a Chloroplast

- Like mitochondria, chloroplasts are double membrane-bound.
- **Stroma:** The space within the inner membrane. Contains enzymes for synthesis of carbohydrates and proteins, small double-stranded circular DNA, and 70S ribosomes.
- **Thylakoids:** Organized, flattened membranous sacs within the stroma.
 - Arranged in stacks called grana (singular: granum).
 - Stroma lamellae are flat tubules connecting the thylakoids of different grana.

KEY CONCEPT FOR NEET: Chlorophyll pigments are present **in the thylakoids**.

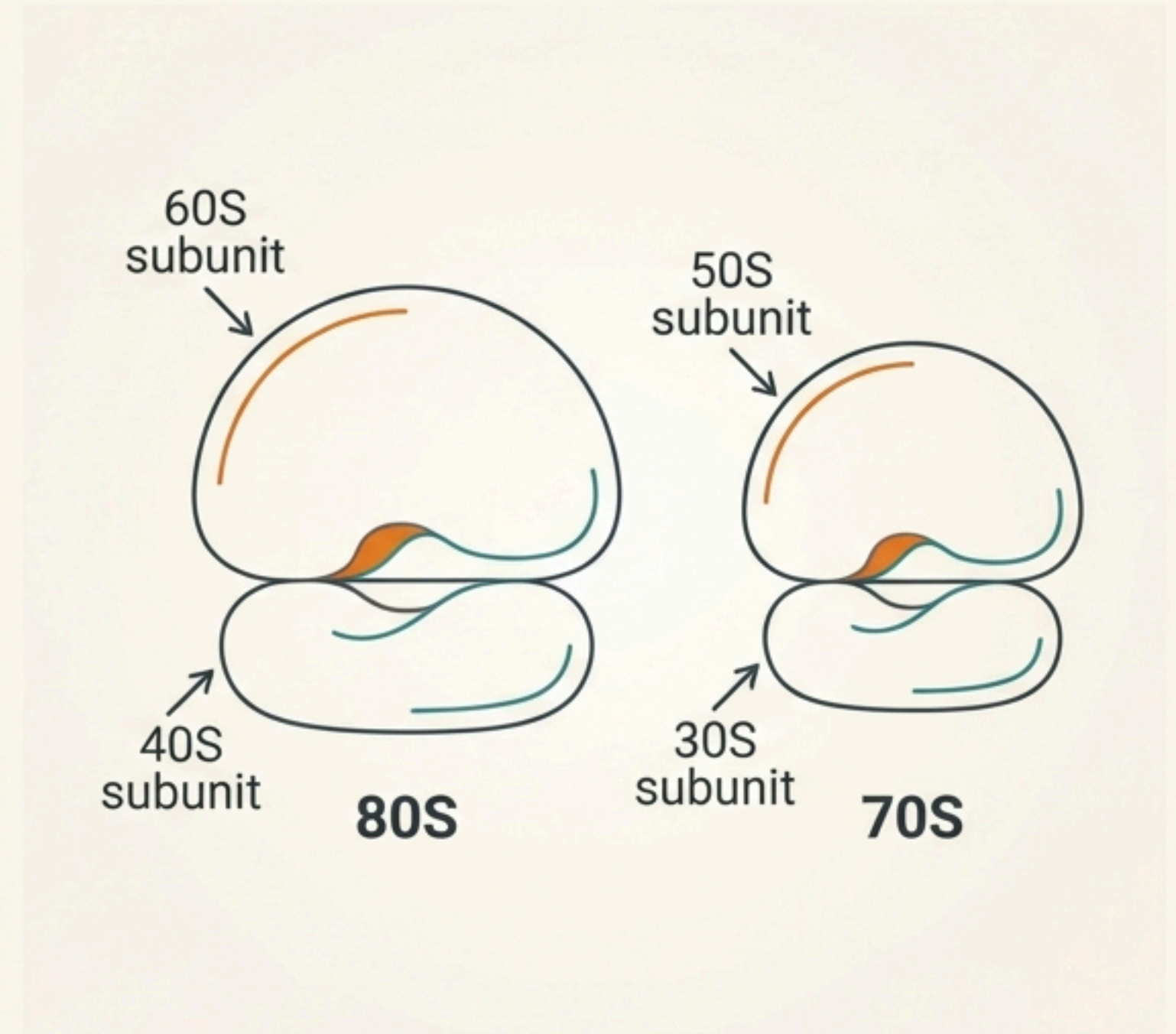


The Protein Factories: Ribosomes

- Granular structures first observed by **George Palade** (1953).
- Composed of **Ribonucleic acid (RNA)** and **proteins**.

KEY CONCEPT FOR NEET: Ribosomes are **not** surrounded by any membrane.

- **Eukaryotic ribosomes are 80S** (composed of 60S and 40S subunits).
- **Prokaryotic ribosomes are 70S** (composed of 50S and 30S subunits). *Note: Found in mitochondria and chloroplasts as well.*
- 'S' stands for **Svedberg's Unit**, a measure of sedimentation coefficient (density and size).



Knowledge Check: Part 1

MCQ Practice for NEET-UG

1. Which of the following organelles is NOT considered a part of the endomembrane system?
a) Golgi complex
b) Mitochondrion
c) Lysosome
d) Endoplasmic Reticulum
2. The synthesis of lipid-like steroidal hormones in animal cells occurs in the:
a) Rough Endoplasmic Reticulum
b) Golgi apparatus
c) Smooth Endoplasmic Reticulum
d) Ribosomes
3. The formation of glycoproteins and glycolipids is an important function of the:
a) Mitochondrion
b) Endoplasmic Reticulum
c) Lysosome
d) Golgi apparatus
4. The membrane surrounding the vacuole in a plant cell is known as the:
a) Tonoplast
b) Plasmalemma
c) Crista
d) Thylakoid
5. The infoldings of the inner mitochondrial membrane are called:
a) Grana
b) Thylakoids
c) Cristae
d) Cisternae

Knowledge Check: Part 2

MCQ Practice for NEET-UG

6. Which of the following is found within the mitochondrial matrix?
- a) 80S ribosomes
 - b) Single, circular DNA molecule
 - c) Stroma
 - d) Glycolytic enzymes
7. Plastids that store carbohydrates (starch) are known as:
- a) Chromoplasts
 - b) Elaioplasts
 - c) Aleurooplasts
 - d) Amyloplast
8. In chloroplasts, the chlorophyll pigments are primarily located in the:
- a) Stroma
 - b) Thylakoids
 - c) Outer membrane
 - d) Stroma lamellae
9. Which of the following is a non-membranous organelle composed of RNA and proteins?
- a) Lysosome
 - b) Ribosome
 - c) Vacuole
 - d) Peroxisome
10. The subunits of a eukaryotic 80S ribosome are:
- a) 50S and 30S
 - b) 60S and 30S
 - c) 50S and 40S
 - d) 60S and 40S

****Answer Key*: 1(b), 2(c), 3(d), 4(a), 5(c), 6(b), 7(d), 8(b), 9(b), 10(d)**